



Surround Yourself With Sound From Three-Channel Audio Amplifier

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This article presents a simple, low-cost audio amplifier with three channels built around two

TEA2025 ICs. The amplifier works from an AC or a DC power supply in the range of 3V to 12V (max. 15V).

This audio amplifier has a left channel, a right channel and one additional channel for the sum of the signals from both channels or for subwoofer or other audio effects.

Approximate Voltage Gains in TEA2025

Table 1

Mode of operation	Gain, dB	Gain, V/V
Gain without external resistor for IC1	43 to 47	140...224
Minimal gain with external resistor for IC1	36	63
Gain in bridge mode	49 to 53	282...447

Description of the circuit

Circuit diagram of the 3-channel audio amplifier is shown in Fig. 1. It is built around two TEA2025 ICs (IC1 and IC2). Each IC has two internal

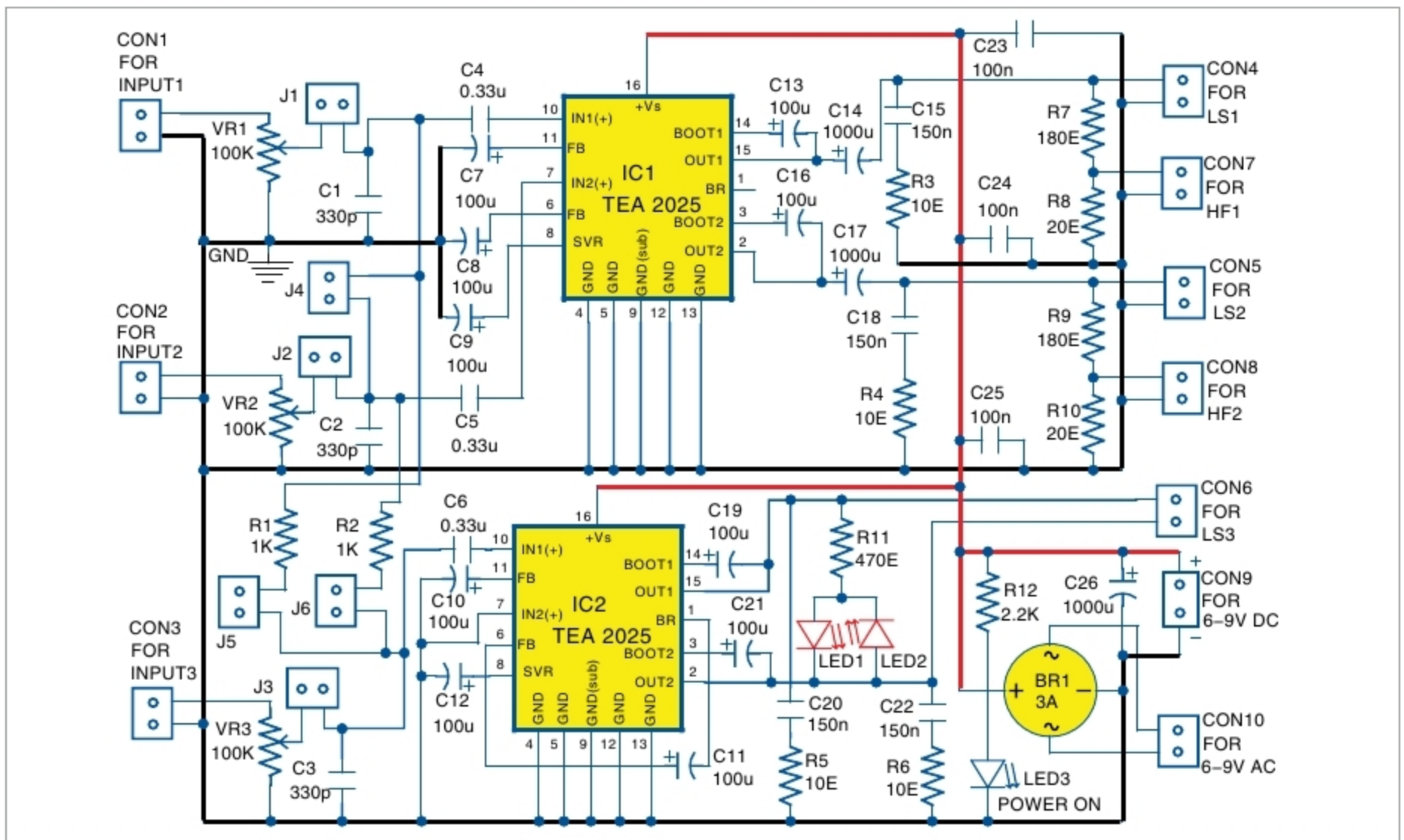


Fig. 1: Circuit diagram of audio amplifier with three channels

Different Configurations and Modes of Operations of The Amplifier

Table 2

Mode	J1	J2	J3	J4	J5	J6
Three separate channels	On	On	On	Off	Off	Off
Stereo and the third channel with the sum of first and second channels	On	On	Off	Off	On	On
First and third channels together, the second channel is alone	On	On	Off	Off	On	Off
Second and third channels together, the first channel is alone	On	On	Off	Off	Off	On
First and second channels together, the third channel is alone	On	Off	On	On	Off	Off